



VIREO @ Video Browser Showdown 2026

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Abstract. In this paper, we report three upgrades to our Interactive Video Search system. (i) A unified vector-database backend enables hybrid dense-sparse retrieval with in-engine filtering and late-fusion re-ranking, improving responsiveness under mixed embedding search and text-based search. (ii) A self-similarity-matrix (SSM) shot segmentation pipeline adapts to both short and long videos, yielding more coherent units for retrieval and browsing. (iii) We generalize the VIREO grid-sketch from colors to an object-sketch: object-by-cell atoms are precomputed offline and combined interactively with the same recommendation shading as the original interface. Across the Vimeo Creative Commons Collection (V3C), Marine Video Kit (MVK), and Laparoscopic Hysterec-tomies (LHE75) datasets, these updates increase search quality while preserving the system’s interactive workflow.

Keywords: SSM-based shot segmentation · object-sketch search · hybrid search

1 Introduction

The Video Browser Showdown (VBS) has established interactive video retrieval as a system-level challenge that rewards fast, hybrid search and effective interaction design on heterogeneous video collections [10, 18]. Community reports highlight three pain points: (i) unifying dense embeddings with lexical signals under tight latency, (ii) supplying coherent browsing units via robust shot segmentation, and (iii) supporting spatially grounded, object-centric queries for known-item and ad-hoc tasks [10, 18].

Building on our system *Interactive Video Search with Multi-modal LLM Video Captioning* [3], we introduce practical upgrades that address these needs. First, we migrate retrieval to a vector-native backend (Milvus) to consolidate dense (e.g. Approximate Nearest Neighbor (ANN)/ Hierarchical Navigable Small World (HNSW)) and sparse (BM25) search with in-engine filtering and late fusion [24]. Second, we replace uniform shot splitting with a self-similarity-matrix (SSM) pipeline that detects novelty, runs kernel temporal segmentation on a candidate-restricted SSM, and refines/merges boundaries to adapt across

short and long videos. Third, we generalize our grid sketch from colors to *object sketch*: object-by-cell atoms are precomputed offline (probability–IoU scoring) and combined online with the same recommendation shading as our previous interface.

In summary, the updated system targets higher retrieval quality and faster interaction while preserving multimodal Visual Language Model (VLM) assistance for query reformulation. The design aligns with recent VBS findings on hybrid retrieval, segmentation quality, and object-centric interaction, and is applied across the V3C [19], MVK [23], and LHE75 [12] datasets.

2 Related Work

Vector Databases for Hybrid Retrieval. The adoption of vector databases has accelerated in interactive video search, enabling unified dense–sparse retrieval with in-engine filtering. VideoEase (VBS 2025) builds its backend on Milvus to index CLIP/BLIP2/OpenCLIP embeddings and support large-scale vector search at interactive rates [22]. The Milvus system itself provides ANN indexing and hybrid query processing for vector data management [24].

Shot Segmentation. Shot boundary detection remains a core pre-processing step in VBS pipelines. Deep architectures such as TransNetV2 deliver accurate, real-time shot transition detection and are frequently integrated by VBS teams [21]. System overviews (e.g., diveXplore at VBS 2024) emphasize the importance of reliable segmentation for downstream retrieval and browsing [20].

Object-centric and Sketch-based Search. Grid-based sketching was popularized in VIREO’s color-grid interaction [13], later extended across VBS editions. Contemporary systems broaden the paradigm to objects and spatial relations: VISIONE encodes object/color presence and spatial constraints via textual indices [1], while IVOS offers object-based filtering by name, location, size, and color [17]. Our object-sketch search follows this trajectory by replacing colors with object-by-cell atoms while preserving pre-computation and fast late fusion.

3 Updates

3.1 Vector Database Updates

We migrated our vector search backend to *MILVUS*, a vector data management system [24], to natively support dense embedding search, sparse/lexical search, scalar filters, and late fusion with re-ranking.

Approximate Nearest Neighbors Search for Dense Embeddings. Milvus exposes multiple ANN indexes for floating-point vectors. We use *graph-based HNSW* and tune index parameters (e.g., graph connectivity, search breadth) to achieve target recall under latency and memory constraints.

Text Search via Sparse Embeddings. We re-indexed the TRECVID dataset using state-of-the-art tools: *PaddleOCR v5* [5] for optical character recognition (OCR) and *OpenAI Whisper-Turbo* [16] for automatic speech recognition (ASR). The extracted OCR texts, ASR transcripts, and associated metadata for each video are indexed using Milvus’s built-in *Sparse-BM25* for text search.

Search Filtering. To introduce custom filters during searching, we utilize the *Metadata Filtering* feature provided by Milvus for this task. With this feature, collections can have scalar metadata fields (e.g., integers, booleans, enums) or more complex attributes. At search time, we attach filter expressions to restrict the subset of entities considered before or during ANN/BM25 search.

Fusion of Ranked Lists. To fuse different ranked lists generated by the different queries (e.g., deep embeddings, text match with BM25, and filters), Milvus enables us to do a simple *linear weighted fusion* between the results.

3.2 Shot Segmentation from a Self-Similarity Matrix (SSM)

As we experienced in Video Browser Showdown 2025, our system performed very well on the V3C dataset but underperformed on both the MVK and LHE75 datasets. The main reason is that we performed uniform shot cutting on these datasets, which leads to two drawbacks: shots are not fine-grained enough for MVK (short videos with a length of 30 s on average), and shots are too densely sampled for LHE75 (long videos with a length of 1 h 20 min on average). To overcome this problem, we propose a simple yet powerful approach to segment video shots for these two video datasets. This approach includes the following steps:

1. **Feature extraction and SSM construction.** We densely sample video frames and extract frame embeddings with BLIP vision-language model [8]. The SSM $K \in [0, 1]^{N \times N}$ is formed by cosine similarity between all pairs of frame embeddings. Before processing, we symmetrize K , set the diagonal to 1, and clip values to $[0, 1]$ for numerical stability. Parameters: sampling rate.
2. **Novelty on the matrix diagonal.** We convert the SSM into a one-dimensional novelty signal by correlating a Gaussian-tapered checkerboard kernel along the main diagonal [6, 11]. When the kernel is centered on a boundary, within-shot blocks align with positive lobes and cross-shot blocks with negative lobes, producing a strong response. One or multiple window sizes may be used (multi-scale). Parameters: novelty window sizes, multi-scale aggregation.
3. **Candidate selection.** We select candidate cut locations as the strongest novelty peaks under a minimum spacing and augment them with a coarse uniform grid to ensure coverage when novelty is weak or slightly misaligned. The union forms the candidate set of potential cuts. Hyper-parameters: number of peaks, peak spacing, grid stride.
4. **Kernel Temporal Segmentation (KTS) on a compressed SSM.** We index K at candidate rows/columns to build a compressed SSM and run

Kernel Temporal Segmentation (KTS) with dynamic programming and an automatic cut penalty [2, 7, 15]. A minimum-length constraint on the compressed axis discourages very short segments. The selected cuts are mapped back to frame indices. Parameters: penalty scale, minimum segment length.

5. **Local refinement.** Each internal cut is refined on the full-resolution SSM by searching a small neighborhood and choosing the index with the weakest cross-boundary similarity. Parameters: refinement radius.
6. **Merging weak boundaries.** To avoid over-segmentation between very similar adjacent shots, we compare average similarity within the left and right shots to the cross-boundary similarity. If the boundary provides little separation - or its novelty is low relative to the overall distribution - we delete the boundary and merge the shots. Block averages are computed efficiently with summed-area (integral) tables [4]. A small number of passes typically suffices. Parameters: absolute/relative thresholds, novelty quantile gate, minimum gap after merging, number of passes.

Parameters. Sampling rate, novelty windows, peak selection and spacing, grid stride, KTS penalty and minimum length, refinement radius, and merge thresholds can be adjusted to control segmentation granularity.

3.3 Efficient Object-Sketch Search

This method is a direct evolution of our grid-based color sketch in [13] and the later sketch interface in [14]. We keep the same ideas: (i) overlay a fixed grid, (ii) precompute and index shot-level responses for all atomic queries, and (iii) support fast ranking and recommendation at query time. The only change is to replace color bins with MS-COCO object classes [9] detected by a strong pretrained model (e.g. Co-DETR/Co-DINO ViT-L [25]). Details are as follows:

Offline precomputation

1. **Frame sampling.** Sample frames from each shot with a configurable rate.
2. **Object detection.** Run a high-accuracy MS-COCO detector to obtain, per frame, detections with class label, segmentation mask, and probability.
3. **Grid overlay.** Place a fixed $H \times W$ grid on the frame.
4. **Atomic query definition.** Every pair (c, g) of object class c (from COCO) and grid cell g is an *atomic query*.
5. **Per-frame cell evidence.** Let $\mathcal{B}_c$ denote the set of all detections for class c on the current frame, each detection $b \in \mathcal{B}_c$ having a confidence $p(b)$ and an object mask. We define $\text{IoU}(b, g)$ as the intersection-over-union between the segmentation of the detection b and the rectangular extent of the grid cell g . The *cell evidence* for (c, g) is

$$s_{c,g} = \begin{cases} \max_{b \in \mathcal{B}_c} (\alpha p(b) + (1 - \alpha) \text{IoU}(b, g)), & \text{if } \exists b \in \mathcal{B}_c \text{ with } \text{IoU}(b, g) > 0, \\ 0, & \text{otherwise,} \end{cases}$$

where $\alpha \in [0, 1]$ balances semantic confidence and geometric alignment. In particular, $s_{c,g} = 0$ when no detection of class c overlaps cell g (including the case $\mathcal{B}_c = \emptyset$).

6. **Shot signature.** Aggregate per-frame evidence to the shot level using max pooling over frames, yielding $S_{\text{shot}}(c, g)$.
7. **Precomputed rank lists for all atomic queries.** For every (c, g) , sort all shots by $S_{\text{shot}}(c, g)$ and store the rank list. Similar to [13], all available queries have a precomputed ranking ready for late fusion and recommendation.

Online query and ranking

We follow the exact interface logic described in [13]:

1. **Select object.** The user chooses an object tag c from a fixed list.
2. **Per-cell recommendation shading.** Immediately after choosing c , the system overlays a recommendation map on the $H \times W$ grid. For each cell g , we estimate the result volume if the atomic query (c, g) were added next. This estimate is computed from the precomputed rank list for (c, g) , conditioned on the atoms already placed [13]. The estimate is normalized to $[0, 1]$ and shown as cell darkness (darker color means more results to examine).
3. **Select grid cell.** The user clicks a cell g , committing the atomic query (c, g) .
4. **Late-fusion update.** The system retrieves the precomputed rank list for (c, g) and fuses it with the running result. The ranked shots refresh instantly.
5. **Iterate.** The user selects the next object; the grid recommendation map is recomputed conditioned on all placed atoms and rendered again. The loop continues until the user is satisfied.

Parameters. Grid size (H, W) , frame sampling rate, α (probability vs. IoU) can be tuned to control granularity and latency.

4 System Overview

The system adopts a two-panel layout: a *browsing* panel on the left and a *query formulation* panel on the right (Fig. 1).

Browsing. The left panel presents ranked retrieval results as a grid of thumbnails. Results are ordered by the current similarity function. Selecting a thumbnail highlights it and enables context actions: retrieve visually similar shots, list other shots from the same video, and play the source video. A secondary view beneath the main grid shows temporally adjacent shots to aid verification and exploration. In the *GPT & SD* tab, users may rewrite the text query with a language model or synthesize a visual proxy to drive content-based search.

Query Formulation. The right panel provides a vertical control panel for composing and constraining queries. A dataset selector switches among collections (e.g., *TRECVID*, *Marine*, *LapGyn*). Visual quality toggles (*BlackBorder*,

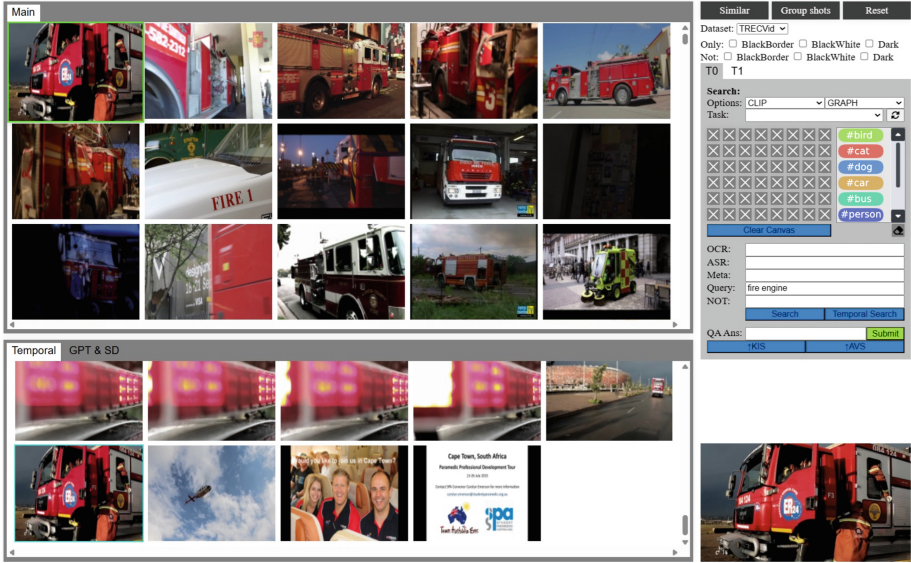


Fig. 1. The system User Interface using *fire engine* as a textual query.

BlackWhite, *Dark*) support both inclusion (*Only*) and exclusion (*Not*) via pre-computed metadata filters. Free-text search is entered in the *Query* field, while modality-specific fields (*OCR*, *ASR*, *Meta*) target text extracted by recognition models or provided by the dataset. Users select the feature index (e.g., *IITV*, *CLIP*, *BLIP*) and the search mode (*graph-based ANN* or *exact cosine*). A 8×6 grid canvas allows spatially grounded visual queries by painting grid cells with colors representing objects. Temporal tabs (*T0*, *T1*) express simple temporal ordering by specifying what should appear first and what should follow. Submissions are task-aware: buttons $\uparrow$ *KIS* and $\uparrow$ *AVS* send results to Known-Item Search or Ad-hoc Video Search, respectively, and a green *Submit* button records a user-provided QA answer.

5 Conclusion

In sum, we unify retrieval on a vector database for hybrid dense-sparse search, adopt adaptive SSM-based shot segmentation, and generalize the color grid to object-by-cell atoms with precomputed ranking and recommendation shading. Together, these updates make interaction faster and results more accurate on the V3C, MVK, and LHE75 datasets, while retaining the system’s interactive workflow and multimodal VLM support.

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